

Algorithm for proving one invariant in one of the branches:

Set up empty proof

1. Add rewrite /operatorname.
2. If there is a forall quantifier
 Add move $\Rightarrow v_1 v_2$ (until) v_k . where k is the number of variables (this can be retrieved by parsing the goal and use the same variable names)
3. If $c = \text{rewrite !get_set_neqE} / \#$. solves the goal
 Add c to the proof and stop
4. While there are updates of maps (look at " $\leftarrow$ " in the goal)
 If $c = \text{rewrite get_set_sameE} / =$. does not result in an error
 Add c to the proof go to next while step
 If $c = \text{rewrite get_set_neqE} / =; 1$: done. does not result in an error
 Add c to the proof go to next while step
 Try rewrite get_setE . and copy the if condition into
 Add case (paste here) $\Rightarrow \text{ieq}$. to proof
 Add + rewrite $\text{ieq get_setE} / =$. to the proof and go back to step 3. using indentation for subgoal proof.
 Go back to step 3.
5. If $c = \text{smt}()$. succeeds:
 Add c to proof and finish algorithm
6. Give up and only keep tactics from step 1. and 2. then admit the goal